

MailMind v2

Production High-Level Architecture

Capgemini AI Buildathon · Use Case #05 · Productivity & Workflow Automation

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★ New in MailMind v2	Purple = AI / LLM layer	Blue = External / Microsoft 365	Teal = Unique v2 features	Amber = Human gate	Green = Confirmed actions
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Zone A — Microsoft 365 / External

MS Graph Webhook new-mail push event	Outlook + Calendar threads · events · To Do	Azure AD / Entra ID OAuth2 · MSAL delegated	Org Graph sender authority rank
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↓ MS Graph webhook push → FastAPI Gateway

Zone B — API Gateway + Auth

FastAPI Gateway /triage · /draft · /approve	Approval Token Gate sendMail blocked until HiL approve	Secret / Key Vault dotenv + Azure Key Vault pattern
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↓ async task dispatch to ingestion queue

Zone C — Ingestion + PII Security (Pipeline Steps 1–2)

Email Ingestion Queue subject · body · sender · thread_id	Presidio PII Masking regex + custom NER → PERSON_n / EMAIL_n	Request Token Map in-process only · never persisted to DB or logs
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↓ masked email payload enters agentic pipeline

Zone D — Agentic Pipeline — Semantic Kernel + LangChain (Steps 3–9)

Semantic Kernel Planner orchestrator loop + memory	LangChain Tool Chain Thread · Calendar · RAG — concurrent fetch	Thread Fetch last 10 msgs via MS Graph
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↓ Steps 3–6 run concurrently via async task group

Commitment Extractor ★ GPT-4o structured JSON · commitments + deadlines	Calendar Conflict Check ★ NER dates vs 72h calendar · badge before open	RAG Retrieval ★ ada-002 · cosine ≥ 0.78 · top-3 precedents
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↓ Steps 7–8: scoring and draft generation

Five-Axis Triage Score ★ 0–100 composite · deadline/authority/sentiment/ thread-age/action · explainable scorecard	Tone DNA Ingestion ★ 180-day sent mail · formality · sentence length greeting patterns · weekly delta refresh	GPT-4o Draft Generation Tone DNA + thread + calendar + RAG context 3 variants: Formal / Balanced / Casual
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↓ Step 9: PII restore before display

PII Restore (Step 9) token map reapplied to draft text	spaCy NER date / entity extraction for conflict check	GPT-4o Classification priority · category · confidence via function call
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↓ assembled payload routed to human gate

Zone E — Human Gate — Mandatory (Step 10) · No bypass code path

React Approval Inbox Next.js 14 · WebSocket draft streaming	Five-Axis Scorecard UI per-bar click-to-explain · conflict badge	Approve / Edit / Reject issues approval token · 3 modes: Gate/Suggest/Review
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↓ approval token required at FastAPI sendMail endpoint

Zone F — Confirmed Actions (Step 11) · Requires approval token

Graph sendMail token-gated at FastAPI layer	MS To Do Task ★ from Commitment Extractor · Confirmation UI	MS Calendar Event ★ deadline write + T-24h proactive alert	Immutable Audit Log who approved · timestamp · edit diff
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↓ writes feed back into data and vector stores

Zone G — Data + Vector Stores

Azure AI Search RAG vector index · PII-masked sent mail alt: ChromaDB / FAISS (free)	Tone DNA Profile Cache stylometric JSON · formality · vocabulary weekly delta refresh via MS Graph	Task Queue / DB async ingestion · audit trail masked content only — no raw PII
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↓ model layer consumes store data at inference time

Zone H — AI Infrastructure

Azure OpenAI GPT-4o chat · extraction · classification function calling for structured JSON	text-embedding-ada-002 Tone DNA ingestion + RAG indexing thread similarity search	Fallback: Groq / Llama 3.1 one env-file switch · zero downtime alt: Ollama local · sentence-transformers
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↓ all model calls instrumented with OTEL spans

Zone I — Observability + Evaluation

OpenTelemetry Spans per-axis triage attributes model · tokens · latency per step	Jaeger Trace judge-linkable URL per email full agent decision path + tool calls	Pytest Golden Dataset 50-item CI harness (5 classes × 10 types) 5 criteria · per-label F1 · confusion matrix
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Pipeline Step Reference

Step	Name	What it does	Technology	v2?
1	Email Ingestion	MS Graph webhook fires on new mail; subject, body, sender, thread_id stored in queue	MS Graph API / FastAPI queue	
2	PII Masking	Presidio (regex + custom NER) replaces PII with PERSON_n / EMAIL_n; token map saved per-request	Microsoft Presidio	
3	Commitment Extract	GPT-4o structured extraction: commitments, action items, deadlines as JSON	GPT-4o function calling	★
4	Calendar Conflict Check	NER extracts dates; compared against 72h calendar fetch; conflict badge computed	spaCy NER + MS Graph	★
5	Thread Fetch	LangChain tool calls MS Graph to retrieve last 10 messages of conversation thread	LangChain + MS Graph	
6	RAG Retrieval	New email embedded with ada-002; top-3 similar past threads retrieved (cosine ≥ 0.78)	ada-002 + Azure AI Search	★
7	Five-Axis Triage	Composite 0–100 score: deadline (30%) · authority (25%) · sentiment (20%) · thread-age (15%) · action (10%)	GPT-4o + org graph + fine-tuned classifier	★
8	Draft Generation	GPT-4o with Tone DNA prefix + masked thread + calendar + RAG precedents; 3 ranked variants	GPT-4o + Tone DNA cache	★
9	PII Restore	Token map applied to re-insert real names/emails into the generated draft before display	FastAPI in-process	
10	Approval Inbox	React UI: five-axis scorecard, conflict badge, precedent citations, 3 draft variants, inline editor	React / Next.js 14 — HUMAN GATE	
11	Confirmed Actions	On Approve: Graph sendMail + MS To Do task + MS Calendar event; immutable audit log written	MS Graph API	★

Tech Stack + Free Alternatives

Layer	Primary (Capgemini Brief)	Free / Open-Source Alternative	Purpose
LLM	Azure OpenAI GPT-4o	Ollama + Llama 3.1 70B / Groq API	Classification · draft gen · commitment extraction
Embeddings	text-embedding-ada-002	sentence-transformers / all-MiniLM-L6-v2	Tone DNA ingestion · RAG indexing · similarity search

Orchestration	Semantic Kernel + LangChain	LangChain alone (OSS) / Haystack 2.0	Planner loop · multi-tool chain binding
Backend	Python FastAPI (async)	FastAPI is already free	Async task groups · approval token gate
Frontend	React / Next.js 14	Streamlit / Gradio	RSC inbox · WebSocket draft streaming · scorecard UI
Vector Store	Azure AI Search	ChromaDB / Qdrant / FAISS (local)	RAG Precedent Engine · PII-masked index
Mail / Cal	Microsoft Graph API	Gmail API + Google Calendar API / IMAP mock	Mail read/send · thread · calendar · To Do
PII	Microsoft Presidio	spaCy + custom regex (local)	Request-scoped masking · custom recognisers
Auth	Azure AD (Entra ID) OAuth2	Auth0 free / Keycloak self-hosted	Delegated OAuth2 · MSAL token refresh
Secrets	dotenv + Azure Key Vault	dotenv alone / HashiCorp Vault OSS	Zero hardcoded keys
Observability	OpenTelemetry + Jaeger + Prometheus	All three are fully open-source	Custom spans · judge-linkable trace URL · metrics
Evaluation	Pytest + 50-item Golden JSON	DeepEval (OSS eval framework)	5 criteria × CI per commit · per-label F1

Production Design Notes

Zero Auto-Send Guarantee	The FastAPI sendMail endpoint requires an approval token that is only issued after a human Approve action in the React UI. There is no code path that routes from Step 9 (PII Restore) to Step 11 (Confirmed Actions) without passing through the Step 10 human gate. This is enforced at the API layer, not the UI layer.
PII Architecture	Microsoft Presidio runs synchronously in-process before any data reaches the LLM, vector index, or log writer. The request-scoped token map is a Python dict — never serialised to disk, cache, or queue. OpenTelemetry traces and Jaeger spans store only masked content. The RAG vector index contains only PII-masked sent mail.
Graceful Degradation	Every unique feature degrades gracefully: RAG falls back to thread-only context if Azure AI Search is unavailable; Tone DNA falls back to balanced tone; Azure OpenAI falls back to Groq/Llama 3.1 via one env-file switch. The demo never crashes due to a dependency failure.
Scalability	The FastAPI backend uses async task groups for concurrent ingestion, embedding, and MS Graph API calls. Long-running operations (Tone DNA ingestion, RAG index build) run as background tasks with UI progress indicators. The vector index supports horizontal scaling via Azure AI Search replicas.
Evaluation CI	A 50-item golden dataset (5 priority classes × 10 email types) runs on every commit via Pytest. Targets: classification accuracy ≥ 94%, draft quality ≥ 88%, context use ≥ 92%, user control 100%, PII protection 100%. Per-label F1 and confusion matrix are visible in the OTel dashboard.

Five unique features. One guarantee: every send is human-approved.
Tone DNA · Five-Axis Triage · Calendar Conflict Detection · Commitment Extractor · RAG Precedent Engine